🔑 **Answer Key — Teacher Copy · Lesson 4-3**

MSTAR PRACTICE WORKSHEET

Lesson 4-3: Estimate the Percent of a Number

Grade 6 Mathematics · Standard 6.AT.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

Margaret's phone has 6 GB of storage and 4 GB is used. About what percent is used?

A. About 25%

B. About 40%

C. About 70%

D. About 150%

4 out of 6 is 2/3, which is about 67%, so 'about 70%' is the reasonable estimate.

Question 2 SELECTED RESPONSE — CORRECT: B

On the double number line for 6 GB, what amount lines up with 25%?

A. 1.25 GB

B. 1.5 GB

C. 2.5 GB

D. 4 GB

25% of 6 GB is one quarter of 6, which is 1.5 GB.

Question 3 SELECTED RESPONSE — CORRECT: A

Which benchmark percents make good landmarks on a number line?

- (A) 0%, 25%, 50%, 75%, 100%
- (B) 3%, 17%, 62%, 94%
- (C) Only 50%
- (D) Only 100%

Multiples of 25 (and 10) are friendly landmarks because their amounts are easy to compute.

Question 4 SELECTED RESPONSE — CORRECT: C

To estimate 8.6% of 216 by rounding, which numbers should you use?

- (A) 1% of 216
- (B) 8% of 300
- (C) 10% of 220
- (D) 86% of 216

8.6% rounds to 10% and 216 rounds to 220 — both are close to the originals and easy to compute.

Question 5 SELECTED RESPONSE — CORRECT: B

What is 10% of 220?

- (A) 2.2
- (B) 22
- (C) 44
- (D) 220

10% as a decimal is 0.1, and 0.1 × 220 = 22.

Question 6 SELECTED RESPONSE — CORRECT: B

The Green-Team recycled 216 pounds; 8.6% was metal. Using compatible numbers 10% and 200, the estimate is:

A 2 pounds

B 20 pounds

C 86 pounds

D 208 pounds

10% of 200 = 0.1 × 200 = 20 pounds of metal.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

A jacket originally priced at $78.99 is on sale for 25% off. Which is the best estimate of the discount amount?

A $8

B $10

C $20

D $40

Round $78.99 to $80. 25% is 1/4 of the whole. 1/4 of $80 is $20.

- **A:** That is about 10% of $80. 25% is two and a half times that much.
- **B:** That is about one eighth of $80. 25% is one QUARTER, so divide $80 by 4.
- **D:** That is half of $80. 25% is half of 50%, so halve it once more.

Part B — correct answer: B

Which benchmark strategy gives the most accurate estimate in Part A?

A Round 25% to 10% and multiply by $78.99.

B Round $78.99 to $80 and find 1/4 of $80 (half of $80 is $40, and half of $40 is $20).

C Round $78.99 to $100 and subtract 25.

D Divide $78.99 by 25.

25% equals 1/4. Finding half of half of a rounded friendly whole ($80) produces an accurate mental benchmark.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, D, E

Select ALL benchmark percent strategies that correctly estimate values:

- **A** To find 10% of any number, divide the number by 10 (move the decimal point one place left).
- **B** To find 5% of a number, find 10% and take half of that value.
- **C** To find 50% of a number, multiply the number by 2.
- **D** To find 20% of a number, find 10% and double it.
- **E** To find 1% of a number, divide the number by 100 (move the decimal point two places left).

Options A, B, D, and E are valid mental benchmark strategies. Option C is incorrect because 50% means dividing by 2 (or multiplying by 0.5), not multiplying by 2.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Write your own estimation problem about percent, using a real situation from your life. Then solve it with a benchmark and say whether your estimate runs high or low.

Model answer: My playlist has 62 songs and about 48% are hip-hop. 48% is about 50%, and 62 is about 60, so about 30 songs. Both roundings moved in opposite directions, so my estimate is very close.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.